



Ollscoil Chathair
Bhaile Átha Cliath
Dublin City University

**School of Computing
RESEARCH ETHICS COMMITTEE**

**APPLICATION FORM FOR ETHICAL REVIEW OF A
RESEARCH PROJECT INVOLVING HUMAN PARTICIPANTS
WHICH IS IN THE CATEGORY OF NOTIFICATION ONLY**

There are 3 generally accepted levels of ethical review for projects carried out in a University or similar setting. These are notification only, expedited and full committee.

This notification only level of review is to approve relatively low-risk research involving human participants, primarily using social science methodologies in which any personal information collected is not of a sensitive nature. The School of Computing Research Ethics Committee has been delegated responsibility by the University to approve ethics submissions from undergraduate and taught Masters projects only, which are in the category of notification only.

Examples of projects in this category include:

- Anonymous surveys in which the topic itself is not likely to elicit significant difficulties for the participants, such as: anonymous internet surveys (e.g. Survey Monkey), street questioning.
- Observation (without audio or visual recording) of public settings where privacy would not normally be expected, such as observing people on streets or at sports events.
- Research carrying no risks beyond those of everyday life (as experienced by the intended participant population), such as asking people's opinions about products or services; asking students about educational experiences; monitoring the impact of daily activities.
- Interviews with public figures, professionals or others in their professional capacity regarding their professional activities.
- Analysis of data (e.g. health records) which have had all identifying information removed by the data holder and been provided to the researcher in accordance with data protection legislation.
- Collection of biological samples which are anonymised and do not require invasive techniques (e.g. hair, nails).

If your project is using data from a public repository like Kaggle or is not generating or using any form of personal data then you do not need research ethics approval, you do not need to complete and to submit this form and your project supervisor should indicate this on the project dashboard.

If your project involves collecting or processing [personal data which is of a personal nature](#), you must first complete the DCU online Data Protection training course and review the ["Data Protection – Key Points for DCU Researchers"](#) guidance from the Data Protection Unit to assist you in meeting your legal obligations under GDPR and associated Irish law.

Once you have completed this form (if you need to) you should save it as a PDF file, not WORD, and upload it to the your project dashboard before you start gathering data. It will then be read and assessed by two members of the committee and once two members of the committee approve your submission you will be automatically notified by email and your project can start data gathering.

There are strict deadlines for submitting this form for each class group, undergraduate and taught Masters by which your submission must be made and you will be informed of these deadlines by your course board chair or project co-ordinator. If you do not submit by these deadlines then the research ethics committee is not obliged to approve your submission and when that happens and your project is assessed and graded at the end of the year, you will be awarded 0 for that component of your project.

SECTION 1 – GENERAL DETAILS**1.1 Project Title**

PulsePath: Festival Crowd Management App

1.2 Applicant Details

Name	Student or Supervisor	E-mail
Zeynep Altintas	Student	Zeynep.altintas2@mail.dcu.ie
Lizel Sulaj	Student	Lizel.sulaj2@mail.dcu.ie
Paul Clarke	Supervisor	Paul.m.clarke@dcu.ie

Other Investigators: *Including any external to DCU*

Name	School/Unit/External Institution	E-mail
N/A	N/A	N/A

1.3 Key Project Dates

Proposed start date for data collection	Proposed end date for data collection	Proposed project completion date
February 2025	April 2025	May 2025

1.4 Please indicate which academic award

Undergraduate ☒	Taught Masters ☐
---	---

1.5 Please confirm the location(s) where the research will be carried out

If research will be carried out abroad, you will need to address the ethical challenges raised by this in Section 3 of your application - consult the Conducting Research Abroad document in the Ethics Resources and Guidelines section of the [DCU Research Ethics webpage](#)).

Dublin, Ireland

1.6 Please state what additional permissions may be required to access participants.

Specify from whom the permission is required (e.g. a school Board of Management), and when their written approval will be obtained

SECTION 2 – PROJECT DESIGN AND METHODOLOGY

Research Overview - Please respect the indicated word counts in the following sections and explain all acronyms in full text the first time they appear.

2.1 Provide a brief description of the research (max 250 words):

Please use lay language, include the scientific/theoretical background of study and a justification as to why this research project should proceed in that context

This research project focuses on the development of a mobile application designed to enhance crowd management at festivals, addressing a need for safety and organisation as emphasised in the Dublin City Council's "Guidelines for Event Organisers". These guidelines lay out the duties of event organisers to ensure that all attendees and staff have a safe and structured environment during events.

Our application aims to facilitate this by providing real-time crowd density data, enabling organisers to monitor crowd movements and implement timely interventions to help minimise overcrowding.

This research is justified within the context of increasing festival crowd management efficiency and the corresponding need for enhanced safety measures. By leveraging technology to address these challenges, our project aligns with contributing to safer and more enjoyable festival experiences.

Ref: Dublin City Council. (2020) Event Guidance

Booklet. https://www.dublincity.ie/sites/default/files/2020-12/dcc_event_guidance_booklet.pdf

2.2 Please state the aims and objectives of the project (max 200 words)

The primary aim of our project is to develop a mobile application that enhances crowd management and improves attendee experiences at festivals and events in Ireland.

Objectives:

Real-time Crowd Monitoring: Implement geolocation and Bluetooth tracking features that provide users with real-time information on crowd density, allowing attendees and organisers to make informed decisions about their movements and resource allocation.

User-Friendly Navigation: Develop intuitive features that can guide attendees to less crowded areas and essential services, such as restrooms and food stalls, thereby promoting safety and comfort.

Data Collection and Analysis: Strive for anonymity when collecting data on crowd movement and density, which can be analysed post-event to improve future planning and safety protocols.

Ref: Health and Safety Executive. (2015). Event safety guide: A guide to health, safety and welfare at music and similar events. Retrieved from HSE Event Safety Guide.

2.3 Please confirm your methods of data collection:

Tick all relevant check boxes and provide details for each one, including any devices used to collect data, and whether the data will be anonymous, potentially identifiable or identifiable at point of collection

Method	Describe briefly
☒ Interviews or focus groups	We will attempt to conduct interviews with festival organisers to gather insights about their experiences and needs regarding crowd management. We will strive for anonymity with any data gathered/taken down on paper.

☒ Surveys/questionnaires	We will conduct surveys or questionnaires for attendees to assess their expectations, experiences, and satisfaction levels with crowd management and event services. We will strive for anonymity with all questionnaire papers filled out.
☐ Audio/video recordings	
☐ Public observations	
☐ Persons in public office	
☒ Using existing data (incl. secondary data)	We will research for any existing data from previous festivals.
☐ Using human derived material (biological samples)	
☐ Standard tests (educational/personality etc.)	
☐ Standard educational practices	
☐ Other (please specify)	

2.4 Please confirm who the participants on this study will be, including group size and composition:

Include associated demographic characteristics, and state how your proposed sample size was determined (e.g. power analysis)

The participants in this study will primarily include festival attendees and possibly festival organisers, with a target sample size of approximately 20-30 individuals.

2.5 Please outline your recruitment process, including where you are sourcing participants from and your criteria for inclusion/exclusion:

Where gatekeepers are involved, outline the procedures relating to their involvement

Our recruitment process involves sourcing participants from an interpersonal connections/ student in our college, focusing primarily on festival attendees who have previously attended large events.

Participants must be over 18 years of age and have experience attending large-scale festivals within the past three years. We will aim to include individuals from different age groups, genders, and backgrounds to reflect the diversity of the attendee population.

We will reach out to festival organisers through contact details we can find online.

2.6 Addressing participant vulnerability – if your participants fall into any of the following categories, please check the relevant tick box/boxes and state below what special arrangements will be made to protect them:

If your participants are not in any of these categories, tick N/A

☒ N/A
☐ Children under 18 years of age
☐ Persons in unequal relationships with the researcher (e.g. lecturer-student, therapist-client, employer-employee)
☐ People with a recognised or diagnosed intellectual, physical or mental impairment

☐ People confined to institutions (e.g. prisoners, residents in 24 hr nursing facilities)
☐ People who have undergone traumatic or adverse emotional events
☐ People with diminished cognitive ability
☐ Marginalised sections of society
☐ Other (please specify)
Special arrangements:

2.7 Involvement of children under 18 years of age – if your participants are in this category, please confirm compliance with the following:

If your participants are not in this category, tick N/A

☒ N/A
☐ We confirm that we have read and agree to act in accordance with the DCU Child Protection policy and procedures (as per the DCU Child Protection Unit webpage)
☐ We confirm that we have put in place safeguards for the children participating in the research
☐ We confirm that we have supports in place for children who may disclose current or historical abuse (whether or not this is the focus of the research)
☐ We confirm that all requirements will be met prior to commencing the research (e.g. TUSLA Children First Training completed, Garda Vetting in place)

2.8 Please confirm how the results of the research will be disseminated:

Include a statement on whether the participants will be provided with any information as to the findings or outcomes of the project

No, the participants will not be provided with any information as to the findings or outcomes of the project.

SECTION 3 – ETHICAL ISSUES AND RISK MANAGEMENT

3.1 Please identify all issues including ethical issues which may arise in the course of this research. What are the potential risks to participants, and how will those risks be addressed or minimised?

Potential risks can be physical, psychological, social, legal, etc. Please include details of any additional support being provided for participants during/after the study

Social Risks: Participants may worry about their responses being linked to their identities. To address this, we will seek to enforce confidentiality and anonymity.
--

3.2 Please identify the potential benefits (direct and/or indirect) to those participating in this research:

Potential benefits should outweigh the potential risks to participants

3.3 Please describe what measures/protocols you have put in place in the event that there are any unexpected outcomes or adverse effects to participants arising from involvement in the research:

- Informed Consent: Before participation, all individuals will receive comprehensive information about the study, including potential risks and benefits through our Informed Consent Form which they will all have to sign, along with a Plain Language Statement.
--

3.4 Do you intend to provide payment or incentives to participants?Yes ☐No ☒

If Yes, please consult the REC Guidelines on the Use of Compensation and Incentives (in the Ethics Resources and Guidelines section of the [DCU Research Ethics webpage](#)) before providing additional details below

3.5 Does this research raise any potential risks for the researchers themselves?

Please consider the location/environment where the research is being conducted, exposure to distressing data content etc.

Yes ☐No ☒

If Yes, please describe further and explain what risk management procedures will be put in place to minimise these risks to researchers:

3.6 Does this research raise any potential conflict of interest?

Please consider any potential real or perceived conflicts of interest that might influence the integrity of the research, or give rise to bias in conducting and reporting the research, or affecting publication (consult the [DCU Conflict of Interest Policy](#) for assistance)

Yes ☐No ☒

If Yes, please identify and explain the steps being taken to address that conflict:

3.7 Please describe how the conduct of the research will be monitored:

Regular oversight by the PI is required to ensure the project conforms to the procedures set out in this application (especially where several people are involved in carrying out the research procedures)

We will hold regular meetings with both team members and the project supervisor to review the progress of the project, discuss any challenges, and ensure that procedures are being followed consistently. These meetings will also provide an opportunity for addressing any deviations from the approved protocol.

SECTION 4 – CONFIDENTIALITY AND DATA MANAGEMENT

4.1 Considering your previous response in section 2.3 of the form on data collection, please confirm whether you are collecting or processing personal data in this research project:

Personal data is any information about a living person, where that person is either identified, or could be identified from the data itself, or when it is combined with other data. This includes paper based, electronic and biological samples data. If your data is fully and completely anonymous, it is not personal data.

Yes ☐	No ☒
------------------------------	--

If Yes, please confirm your compliance with the following by ticking the checkboxes:

☐ We confirm that we have completed the DCU Data Protection training module on Loop.
☐ We confirm that we have read the “Data Protection – Key Points for DCU Researchers” guidance on the DCU Data Protection Unit (DPU) website and agree to protect and manage our data in accordance with same.
☐ We have assessed the degree of risk inherent in the personal data being used in the research project, and confirm that all DPU GDPR requirements have been met prior to submitting this application (e.g. completion of Data Protection questionnaire, confirmation that any survey tool being used is GDPR compliant, that required Data Processing or Sharing Agreements will be in place, etc.)

4.2 Data access – please confirm whether access to participant data is confined to the investigators named on this application:

Yes ☒	No ☐
---	-----------------------------

If No, please name who the other individuals are and why they need access. Any proposed transfer of data (including outside of the EU) should be detailed here.

--

4.3 Data storage – please confirm compliance with the following:

☐ Data collected on mobile devices will be protected with a strong password/passphrase at a minimum, and/or encrypted if the device supports it
☐ Data will be removed from mobile devices as soon as is practicable and stored in a secured location in DCU (on server or institutional Google Drive)
☒ Paper based data will be held securely in locked cabinets in DCU, with access restricted to the named researchers
<u>Specific arrangements in relation to biological samples should be stated here:</u>
<u>Any exemptions to the above compliance statements should be justified here:</u>

4.4 Please confirm who will be responsible for the secure storage of data generated by the research:

Name the relevant DCU investigator/s

Zeynep Altintas
Lizel Sulaj

4.5 Please confirm how long the data will be held for:

For personal data, consult section 15: Retention of Personal Data in the [“Data Protection – Key Points for DCU Researchers”](#) guidance on the DCU Data Protection Unit (DPU) website

The data collected for this research will be retained until mid-May 2025, which aligns with the project completion date.

4.6 Please confirm what will happen to the data collected at the end of the study:

Please tick the relevant checkbox and complete the associated follow-up section for that category

Archived ☐	Destroyed ☒	Other ☐
-----------------------------------	---	--------------------------------

4.6.1 Archived data

Please provide the following details:

Name the DCU staff member responsible for archival and future use of data	
Confirm whether the data will be made available to other researchers, and if so, how?	
Confirm <u>how</u> the data will be prepared for archive (e.g. will datasets be anonymised)	
Confirm <u>where</u> the data will be archived and who will be allowed to access it	

4.6.2 Destroyed data

Please provide the following details – Note: for student projects, the supervisor must take responsibility for data destruction if there is no guarantee the student will have access to the data at the time of destruction

Please justify why the data will be destroyed	Won't be of use to us anymore
Name the DCU researcher responsible for destruction of data	Lizel Sulaj, Zeynep Altintas
Confirm when the data will be destroyed (specify date)	15 th September
Confirm compliance with the following destruction methods (tick relevant boxes)	☐ Electronic data will be overwritten/securely deleted ☒ Paper based data will be confidentially shredded ☐ Medical samples will be disposed in accordance with the relevant DCU approved SOP

4.6.2 Other - Please explain what will happen to the data if not being archived or destroyed:

SECTION 5 – PARTICIPANT INFORMATION AND INFORMED CONSENT PROCEDURES

In addition to completing this form you are required to attach, within the single PDF that you submit, a copy of (1) the Participation Information Sheet which you share with your participants and (2) a copy of the Informed Consent Form which your participants sign.

5.1 Please confirm that the following items have been addressed in your Participant Information Sheet which should be shared with all participants whether it involves online or in-person data gathering:

The items below should be used as headings in your information sheet. Note the language used under each item must reflect the participant age group and corresponding comprehension level– if your participants have different comprehension levels (e.g. both adult and child participants) then separate sheets must be prepared for each set. Templates are available via the [REC Forms - Applications, Templates and Amendments section](#) of the Research Ethics website.

Checklist – tick the relevant check box for each item	Yes	No
Introductory Statement (Researcher names and titles, school, title of the research study)	☒	☐
What is this research about?	☒	☐
Why is this research being conducted?	☒	☐
Why have you been invited to take part?	☒	☐
What will happen if you decide to take part in this research study?	☒	☐
How will your data be used?	☒	☐
How will your privacy be protected (including any legal limits to confidentiality)?	☒	☐
What are the benefits of taking part in this research study?	☒	☐
What are the risks of taking part in this research study?	☒	☐
Can you change your mind at any stage and withdraw from this study?	☒	☐
How will you find out what happens with this project?	☒	☐
Contact details for further information	☒	☐

If you marked any item as No, please explain and justify why:

5.2 Informed Consent Procedures – please confirm whether written consent is to be obtained:

Please tick the relevant checkbox

Yes ☒	No ☐
---	-----------------------------

If Yes, describe the procedures by which written consent will be obtained. If you are involving child participants, you will also need to obtain their written assent. Templates are available via the [REC Forms - Applications, Templates and Amendments section](#) of the Research Ethics website.

We will obtain consent through an Informed Consent Form we will make our participants fill out.

If No, describe the procedures regarding how consent/assent will be obtained:

If you are gathering data from an online process such as Google Form or SurveyMonkey then you should use a page such as the one below, to capture participants' informed consent and your data gathering should not proceed until participants have completed this form with the appropriate answers.

Participant – please complete the following (by clicking Yes/No for each question)

I have read the Plain Language Statement (or had it read to me) *

- ☒ Yes
☐ No

I understand the information provided *

- ☐ Yes
☐ No

I have had an opportunity to ask questions and discuss this study *

- ☐ Yes
☐ No

I understand the information provided in relation to data protection *

- ☐ Yes
☐ No

I have received satisfactory answers to all my questions *

- ☐ Yes
☐ No

I understand I may withdraw from the Research Study at any point *

- ☐ Yes
☐ No

I have read and understand the arrangements to be made to protect confidentiality of data, including that confidentiality of information provided is subject to legal limitations *

- ☐ Yes
☐ No

I have read and understand confirmations relating to any other relevant information as indicated in the PLS *

- ☐ Yes
☐ No

I consent to participate in this research study *

- ☐ Yes
☐ No

SECTION 6 – SUBMISSION CHECKLIST AND RESEARCHER DECLARATION

6.1 Please confirm all required supplementary documentation to be included in this application within Section 7:

Checklist – tick the relevant check box for each item	Yes	N/A
Participant Information Sheet/s	☒	
Informed Consent Form/s	☒	☐
Informed Assent Form/s	☐	☐
Recruitment Advertisement	☐	☐
Questionnaire/Survey	☒	☐
Interview/Focus Group Questions	☒	☐
Debriefing Material	☐	☐
Bibliography	☐	☐
Approval from another Research Ethics Committee	☐	☐
Evidence of other external approvals (e.g. Board of Management letter)	☐	☐
Evidence of internal approvals (e.g. BSC approval review letter)	☐	☐
Other – provide details here:	☐	☐

6.2 Signed Declaration

By submitting this form, the applicant (and supervisor) agree to the following:

The information contained herein is, to the best of my knowledge and belief, accurate. I have read the University's current research ethics guidelines, and accept responsibility for the conduct of the procedures set out in the attached application in accordance with the form guidelines, the [REC guidance and resources](#), the University's [Conflict of Interest Policy](#), its [Code of Good Research Practice](#) and any other condition laid down by the Dublin City University Research Ethics Committee. I have attempted to identify all risks related to the research that may arise in conducting this research and acknowledge my obligations and the rights of the participants.

I also acknowledge my requirement to be informed as to other duties and legal obligations applying to my research, and to comply with these duties and obligations – this includes being informed about DCU Data Protection guidelines for researchers, DCU Child Protection policy and procedures (where relevant) and DCU Insurance requirements.

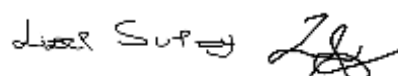
I and my co-investigators and/or supporting staff have the appropriate qualifications, experience and facilities to conduct the research set out in the attached application and to deal with any emergencies and contingencies related to the research that may arise. Research will not commence until required consents and approvals are in place.

Electronic Signature(s):

Supervisor: Paul Clarke

Print Name Here: Paul Clarke

Date: 31/10/2024

Student(s) signature(s): 

Print Name(s) here: Lizel Sulaj Zeynep Altintas

Date: 17/10/2024

SECTION 7 – SUPPLEMENTARY DOCUMENTATION

Please attach all required documentation as confirmed by you in the previous section. The application should then be saved as one file in PDF format before submission via the project dashboard.

Plain Language Statement

Festival Crowd Management: Understanding Festival Attendees' experiences for App Development

Lizel Sulaj, Final Year Student, Computing for Business

Zeynep Altintas, Final Year Student, Computing for Business

School of Computing and Engineering
Dublin City University (DCU)

Who Are We and What is This Study About?

We are Lizel and Zeynep, final-year students at Dublin City University. Our research project focuses on improving festival experiences for attendees by developing a mobile app that helps manage crowds. We are conducting this study to learn more about how festival-goers feel about crowds, how they navigate festivals, and what could be done to make these events more enjoyable.

Why Have You Been Asked to Take Part?

You have been invited to participate in this study because you are a festival-goer, and your experiences and insights are valuable for understanding how crowds impact the festival experience. We are looking for people like you to share their thoughts on how festivals can be improved when it comes to crowded areas.

What Will You Be Asked to Do?

If you agree to participate, you will be asked to complete a short survey that takes about 5-10 minutes. The survey will include questions about:

- Your experience at festivals
- How you handle crowded spaces at festivals
- Your thoughts on what would make your experience better

Your answers will help us design a mobile app that can help festival-goers avoid overcrowded areas and make their time at festivals more enjoyable.

Do You Have to Take Part?

No, your participation is completely voluntary. You are free to choose whether or not you want to take part in this study. If you decide to participate but later change your mind, you can stop at any time without any consequences. There is no penalty for not taking part or for withdrawing at any point.

Will Your Participation be Confidential?

Yes, your participation will be kept completely confidential. We will not collect any personal information that could identify you, such as your name or contact details. All your answers will be anonymous and stored securely. Only the research team will have access to the data, and the results will be used only for this study.

What Are the Possible Benefits of Taking Part?

By participating, you will be helping us understand what makes the festival experience better for attendees. While you may not receive a direct benefit, your input will contribute to the development of a mobile app that could help future festival-goers enjoy festivals more, by avoiding overcrowded areas and managing their time better.

What Are the Possible Risks of Taking Part?

There are no known risks associated with participating in this study. If you feel uncomfortable with any of the questions, you can skip them or choose to stop participating at any time without any impact.

What Will Happen to the Information You Give Us?

The data you provide will be used to analyse festival attendees' experiences with crowds. Your information will be kept anonymous and stored securely on a password-protected database. After the study is completed, the data will be kept until mid-September 2025 and then destroyed. The results of the study may be published in academic papers, presented at conferences, shared with other researchers or potential customers (festival organisers) but no personal details will be shared, and you will not be identified.

How Will You Find Out What Happens with This Study?

If you are interested in knowing the results of the study, we can send you a summary once the research is completed. You can provide your email address at the end of the survey if you wish to receive this. Your email will be used solely for this purpose and will not be linked to your survey responses.

Who Can You Contact for More Information?

If you have any questions about the study or your participation, please feel free to contact us:

Lizel Sulaj
lizel.sulaj2@mail.dcu.ie
(083) 002 8851

Zeynep Altintas
zeynep.altintas2@mail.dcu.ie
(087) 168 7370

Thank you for considering participating in our study!

Participation Information Sheet

Festival Crowd Management: Understanding Festival Attendees' experiences for App Development

Lizel Sulaj, Final Year Student, Computing for Business
Zeynep Altintas, Final Year Student, Computing for Business

School of Computing and Engineering
Dublin City University (DCU)

This research is being conducted as part of our final-year project at Dublin City University. Our study aims to gather insights from festival-goers about their experiences in managing crowded areas, navigating festival grounds, and overall satisfaction with their festival experience. The data collected

will be used to inform the development of a mobile app that aims to enhance crowd management at festivals, improving safety and attendee enjoyment.

Why is this research being conducted?

You have been invited to participate because you are someone who has attended or will attend festivals. Your insights and experiences are essential for us to understand what festival-goers need and want.

We are specifically looking to gather insights from a diverse group of attendees to ensure that our findings reflect a wide range of experiences and preferences.

If you are a student or staff member at Dublin City University, please note that your participation in this study is entirely voluntary. Your involvement or non-involvement in this study will not affect your ongoing assessment, grades, or any professional relationships you have with the researchers. Participation is independent of any academic or administrative evaluations.

What you have to do

If you choose to participate, we will ask you to fill out a survey. The survey will include questions about:

- Your experiences at festivals, especially regarding crowded areas
- How you find your way around festivals
- What could be done to make your experience better

The survey should take about 5-10 minutes to complete. You can answer the questions online or on paper or in person.

How your data will be used

We will analyse the collected data to identify common themes and insights related to crowd management and festival experiences. This analysis will help us understand what festival-goers like you need in a crowd management app. The results of the study may be presented in academic reports, presentations, and publications. However, any information shared we will strive for it to be anonymized. All data will be stored securely, with access restricted to the named researchers. The data collected for this research will be retained until mid-September 2025.

You can withdraw your data at any time during the study. If you choose to withdraw, please contact us, and we will remove your information from our database.

DCU Data Protection Officer – Mr. Martin Ward (data.protection@dcu.ie Ph.: 7005118 / 7008257) will handle any data protection concerns arising from the research.

You have a right to lodge a complaint concerning the use of personal data with the [Irish Data Protection Commission](#) if deemed necessary.

This data might later be anonymised (e.g. for statistical or aggregated data) to share with potential customers (festival organisers).

Please note that confidentiality of information can only be protected within the limitations of the law - i.e., it is possible for data to be subject to subpoena, freedom of information claim or mandated reporting by some professions

What are the benefits?

As far as the pros for taking part; while you may not receive direct benefits, your input will help us create a better festival experience for everyone. Your opinions can lead to the development of an app that helps attendees avoid crowded areas and enjoy festivals more.

What are the risks?

There are no known risks involved in this study. You can skip any questions you do not want to answer, and your participation will remain confidential.

Can you change your mind at any stage and withdraw from this study?

Participation in this study is entirely voluntary, and will end at the point you decide to withdraw your consent to participate. Your data can be removed from our database at any time up to the point of anonymisation or publication.

If you're looking to withdraw your data please contact either of the following; Lizel Sulaj (lizel.sulaj@mail.dcu.ie, (083) 002 8851), Zeynep Altintas (zeynep.altintas2@mail.dcu.ie, (087) 168 7370).

How will you find out what happens with this project?

At the end of the study, we will share a summary of the key findings with all participants who are interested in knowing the results. This summary will include insights from the data we collect about festival experiences and how it informed the development of our crowd management app.

If you would like to receive this summary, you can provide your email address at the end of the survey. Your email address will only be used to send you the study's results and will not be linked to your survey responses, ensuring your anonymity.

If participants have concerns about this study and wish to contact an independent person, please contact:

The Secretary, Dublin City University Research Ethics Committee, c/o Research and Innovation Support, Dublin City University, Dublin 9. Tel 01-7008000, e-mail rec@dcu.ie

Informed Consent Form

Festival Experience Survey for Festival Crowd Management App Development

Investigators: Lizel Sulaj, Zeynep Altintas
School of Computing and Engineering
Dublin City University

You are invited to participate in a research study conducted by Lizel Sulaj and Zeynep Altintas as part of our final-year college project. The purpose of this study is to gather insights about the festival experience of attendees, including factors related to crowding, navigation, and overall satisfaction. This information will help us in designing a crowd management mobile app to improve festival experiences.

If you agree to participate, you will be asked to complete a short survey. The survey will ask questions about:

- Your experiences with crowds at festivals
- How you navigate during festivals
- Preferences for managing crowded areas

The survey should take approximately 5-10 minutes to complete.

Your participation in this study is *entirely voluntary*. You have the right to refuse to participate or to withdraw at any point during the survey, without any penalty or loss of benefits.
Confirmation of arrangements to be made to protect confidentiality of data, including that confidentiality of information provided is subject to legal limitations.

Please complete the following:

I have read the Plain Language Statement (or had it read to me)	Yes No
I understand the information provided	Yes No
I have had the opportunity to ask questions and discuss this study	Yes No
I have received satisfactory answers to all my questions	Yes No
I am aware that my interview will be audiotaped	Yes No

I consent to the use of my data for future studies within the following parameters:

The information collected from this survey may be used in academic presentations, reports, or publications. However, no personally identifiable information will be used, ensuring your privacy and anonymity.

We will strive for the data to be anonymised to ensure no individual identities can be identified.

The data will be destroyed after the completion of this project in mid-September 2025.

I have read and understood the information in this form. My questions and concerns have been answered by the researchers, and I have a copy of this consent form. Therefore, I consent to take part in the research project

Participants Signature:

Name in Block Capitals:

Witness:

Date:

Festival Crowd Management Questionnaire

1. How many festivals have you attended in the past 12 months?

☐

0

☐

1-2

☐

3-5

☐

More than 5

2. What kind of festivals do you usually attend?
(Select all that apply)

☐

Music festivals

☐

Food and drink festivals

☐

Arts and cultural festivals

☐

Other (please specify)

3. How would you rate your overall satisfaction with the festivals you've attended?

☐

Very satisfied

☐

Satisfied

☐

Neutral

☐

Dissatisfied

☐

Very dissatisfied

4. How do you usually navigate festival grounds?
(Select all that apply)

☐

Festival-provided map (app or paper)

☐

Asking for directions

☐

Following signage

☐

Walking around until I find my destination

☐

Other (please specify)

5. How difficult is it for you to find less crowded areas during festivals?

☐

Very easy

☐

Easy

☐

Neutral

☐

Difficult

☐

Very difficult

6. Have you ever missed a scheduled event (e.g., performance, activity) because you were stuck in a crowded area?

☐

Yes

☐

No

7. What areas of the festival do you find are most crowded?
(Select all that apply)

☐

Entry/exit points

☐

Food and drink stalls

☐

Main stage or performance area

☐

Restrooms

☐

Merchandise stalls

☐

Other (please specify)

8. Have you ever felt unsafe due to overcrowding at a festival?

☐

Yes

☐

No

9. Would you prefer to avoid certain areas if you knew they were overcrowded?

☐

Yes

☐

No

10. What is your age group?

☐

Under 18

☐

18-24

☐

25-34

☐

35-44

☐

45 and above

11. Do you have any accessibility needs that affect how you move around at festivals?

☐

Yes

☐

No

If "Yes" Please describe:

Thank you for completing our questionnaire!

Interview Questions for Festival Organisers

1. Can you describe the main challenges you face in managing large crowds at your festivals?
2. What systems or tools do you currently use to monitor crowd flow and capacity at various festival areas?
3. How do you manage safety and security concerns, especially in high-traffic or overcrowded areas?
4. How do you measure and track the success of the festival in terms of crowd control and attendee satisfaction?
8. How do you currently communicate important information (schedule changes, area closures, overcrowding) to festival attendees?
9. What are key legal or regulatory concerns when it comes to crowd control and safety at your festival?
10. How do you handle attendee data privacy and ensure compliance with data protection laws?
11. What feedback do you usually receive from festival-goers regarding crowding and ease of navigation?
12. How do you manage vendor locations and ensure they are distributed to avoid overcrowding in particular areas?
13. How do you coordinate with event vendors and staff to handle high-traffic areas during peak times?
14. What are your long-term goals for improving crowd management and the overall festival experience?

